VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANA SANGAMA, BELAGAVI– 590018, KARNATAKA, INDIA

A PROJECT REPORT
on

ADVERSARIAL ROBUSTNESS EVALUATION AND DEFENSIVE FRAMEWORK FOR DEEP NEURAL NETWORK

Submitted in partial fulfilment of the requirements for the award of

BACHELOR OF ENGINEERING

in

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

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VIVEKANANDA COLLEGE OF ENGINEERING & TECHNOLOGY
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DEPARTMENT OF ARTIFICIAL ENGINEERING & MACHINE LEARNING

CERTIFICATE

Certified that the project work entitled **Adversarial Robustness Evaluation and Defensive Framework for Deep Neural Network** is carried out by **Mr. Deepthik Rai B, Mr. Pavan Kumar HP, Mr. Dhruvanarayana S, Mr. Tanmay SS** bearing USNs **4VP23AI014**, **4VP23AI031, 4VP23AI017 and 4VP23AI058** respectively bonafide students of **Vivekananda College of Engineering & Technology, Puttur** in partial fulfilment for the award of **Bachelor of Engineering** in **Artificial Intelligence & Machine Learning** of the **Visvesvaraya Technological University, Belagavi** during the year 2025-26. It is certified that all corrections/suggestions indicated during Internal Assessment have been incorporated in the report deposited in the departmental library.

The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

Signature of the Guide	Signature of Project Coordinator	Signature of the HOD
Prof. Vaishnavi KV	**Prof. Ajay Shastry C G**	**Prof. Abhishek Kumar K**

DECLARATION

We, **Mr. Deepthik Rai B (4VP23AI014), Mr. Pavan Kumar HP (4VP23AI031), Mr. Dhruvanarayana S (4VP23AI017), Mr. Tanmay SS (4VP23AI058)** students of B.E. 6th Semester in Artificial Intelligence & Machine Learning, **Vivekananda College of Engineering & Technology**, Puttur, hereby declare that the project work entitled **Adversarial Robustness Evaluation and Defensive Framework for Deep Neural Network** has been carried out by us at VCET, Puttur, under the guidance of **Prof. Vaishnavi KV,** Assistant Professor, Department of Artificial Intelligence & Machine Learning, Vivekananda College of Engineering & Technology, Puttur, and submitted in partial fulfilment of the requirements for the award of degree in **Bachelor of Engineering in Artificial Intelligence & Machine Learning** by **Visvesvaraya Technological University**, Belagavi during the academic year 2025-2026.

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ABSTRACT

Deep neural networks (DNNs) represent an essential tool in today's AI industry and contribute to image recognition, natural language processing, autonomous technology, etc. Although DNNs work excellently in many cases, they suffer from adversarial attacks. As a rule, the problem is connected to the vulnerability of the model towards small perturbations in input data leading to the wrong output. Popular methods of attacking DNNs include, among others, Fast Gradient Sign Method (FGSM) and Projected Gradient Descent method (PGD).

In case of ignoring this issue, the reliability of the system suffers greatly. The majority of today's methods for evaluating deep learning models do not go beyond traditional test methodologies failing to detect the impact of adversarial attacks. The aim of the current project is to develop a system that will help to improve the evaluation process and enhance robustness of the deep learning model.

The core element of the system is represented by its testing component that allows testing a pre-trained deep learning model by applying different kinds of adversarial attacks to the given dataset. Moreover, the model can be tested both in normal mode and in adversarial mode to see the difference in the model's performance.

It should be noted that testing module works in connection with simple web interface based on tools like Streamlit. The user can provide information on his/her needs, enter necessary parameters of input, and view the results of predicting, as well as the change of output when it comes to applying adversarial noise. In addition, the system provides recommendations on defensive techniques.

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LIST OF ABBREVIATIONS

Abbreviation	Description
CNN	Convolutional Neural Network
MFCCs	Mel-Frequency Cepstral Coefficients
ROC curve	Receiver Operating Characteristic curve
ReLU	Rectified Linear Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
SVM	Support Vector Machine
R-CNN	Region Based CNN
GUI	Graphical User Interface
YOLOv5	You Only Look Once version 5
MDSS	Medical Decision Support System
MCC	Matthews Correlation Coefficient
ANN	Artificial Neural Network
CLAHE	Contrast Limited Adaptive Histogram Equalization
Grad-CAM	Gradient-weighted Class Activation Mapping
RNN	Recurrent Neural Networks
NumPy	Numerical Python
LSTM	Long Short-Term Memory

CHAPTER 1

INTRODUCTION

1.1 Introduction to the Project

Deep Neural Networks (DNNs) have become prevalent components in a vast array of artificial intelligent systems and applications, such as image classification, natural language processing and autonomous vehicles, where their state-of-the-art performance is often heavily relied upon. However, a pervasive threat to these networks is various forms of adversarial attacks.

Detecting the vulnerabilities of machine learning models early is key to avoiding disastrous errors. Standard methods for model evaluation are typically inadequate and may even introduce errors or fail in certain circumstances. Standard testing techniques often fail to handle and reveal how models behave under stress and attacks. With the ever growing field of artificial intelligence, and the rapidly emerging deep learning security area, there is a strong need to evaluate the models' robustness against adversarial attacks and defend them from such attacks.

The goal of this project is to implement an intelligent system that autonomously tests and enhances the robustness of deep learning networks using adversarial techniques. We apply methods to generate adversarial examples (FGSM, PGD) to trained models and aim to detect vulnerabilities of the model's predictions in perturbed regions of the input space. In addition, basic defense mechanisms should be implemented to evaluate their effectiveness. The system will be finished and provided as an user-friendly interface where one can upload images (or other data) and see how the model reacts during normal operation as well as under adversarial distortion.

The goal of this work is to help researchers and developers better assess the limitations of current deep learning models and to improve their reliability.

1.2 Existing System

In recent years, Deep learning models have been gaining increasing popularity in recent years, and tools and techniques to evaluate their performance and security have been developed as well. Most of such methods are based on common benchmarks, basic robustness tests, and simple attacks. Although some frameworks and libraries, such as TensorFlow and PyTorch, support adversarial testing, existing approaches are not universally effective and lack compatibility for various types of deep neural networks.

Most existing systems are designed for broad applicability but lack effectiveness when it comes to in-depth model analysis under strong adversarial attacks. Typically, such systems require many configurations and considerable computational resources, which often renders them inaccessible to students and hobbyists without robust hardware who are merely interested in gaining insight into model robustness.

A significant gap currently exists between how models are evaluated in comparison to how they are challenged by adversarial attacks. Currently, model evaluation relies on standard accuracy metrics and insufficient forms of validation, providing little insight into the performance of models under attack. Proper analysis and visualization of results are therefore required to better identify weaknesses and improve the security of current models.

Due to the complexity, unreliability, inefficiency, and unfeasibility of current methods for evaluating the robustness of deep neural networks, there is a significant need for a practical, simple, and manageable system that leverages advanced adversarial methods to improve the robustness of deep learning models by providing a comprehensive analysis post-defense.

1.3 Proposed System

We propose a system for robustness evaluation and improvement of deep neural networks based on adversarial learning. To evaluate robustness of a deep learning model, we employ deep learning models and state-of-the-art adversarial attack techniques that investigate how small perturbations of input samples affect the behavior of a model.Starting from a given dataset and a trained deep learning model for classification tasks, the system generates inputs for the model's evaluations, normalizes and formats them, and then applies adversarial attacks such as FGSM and PGD to them. The system is implemented using TensorFlow and Keras and tests the model under normal and adversarial conditions. After the evaluation, the system compares the predictions given by the model to those under normal conditions and computes the drop in accuracy to quantify the model's vulnerabilities.

Additionally, the system has some basic defense mechanisms like adversarial training and some input preprocessing techniques. This is part of the final system where you can input data samples (either type them in or upload them as a file) and the system shows the results of both normal and the adversarial processing for a given input. Results include the predicted values, the confidence in the results, and a performance comparison chart.

To make our system more user-friendly and easily accessible, we built a simple Streamlit interface that allows users such as students, researchers, or other developers to interact with our system in real-time. They can see the results of how adversarial attacks decrease the accuracy of a model and how defensive techniques can improve a model's performance.

We present a novel system that combines adversarial evaluation with an intuitive interface for practical, reliable and efficient deep learning model analysis and improvement.

1.4 Scope of the Project

In this project we attempt to add robustness to existing deep neural networks, and explore their behaviour using adversarial methods. We study the input to a model in order to inform the researcher/developer of vulnerabilities to the FGSM and PGD attacks. A deep learning based implementation of the model has been created and surrounded by a simple web-based interface using the Streamlit framework. Users can either type in input directly in the interface or upload files to the server for analysis. The model’s behaviour under normal and under attack conditions is computed, classified and returned along with a confidence score. For vulnerable inputs, information on existing defence mechanisms and possible improvements to the model are also provided. The goal of this project is to deliver an interactive, efficient and simple to use solution, which gives enough insight into a model’s behaviour for users without an in-depth technical background. Intelligent evaluation and corresponding defence techniques are used to improve the reliability and security of existing deep learning models.

1.5 Objectives of the Project

The primary goal of this project is to create a robust, reliable, and accessible system for utilizing adversarial learning techniques to improve the robustness of deep neural networks. To pursue this goal, we first establish a set of more specific objectives that need to be satisfied. These goals are targeted at identifying techniques to analyze and defend models, increasing usability and performance, providing real-time or near real-time feedback, and supporting the developers of AI systems.

An intelligent framework is developed using deep learning and adversarial learning techniques to evaluate whether a machine learning model performs correctly under normal conditions and remains robust against adversarial attacks such as Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD).

The system analyzes both standard and adversarial inputs to identify vulnerabilities and assess the reliability of model predictions. A user-friendly web-based interface is provided to simplify data input, prediction analysis, and visualization of confidence scores and comparative results.

The framework also assists developers in identifying weaknesses in the model and suggests suitable defensive mechanisms to improve robustness and security against adversarial manipulations.

1.6 Organization of the Report

The report is organised in chapters that describe the individual elements of the project in a logical and coherent order, explaining the process of identifying the problem, working on a solution and assessing the results.

Chapter 1 This document begins by introducing the project and outlining the core objective of the work, which is to develop a system which analyses and improves the robustness of Deep Neural Networks using Adversarial approaches. The document goes on to detail why such work is important to computing as a whole, particularly within those fields in which the reliability and security of models is paramount to their effective use. The motivation for the project is then followed by the objectives of the work as well as a plan of how the document will progress throughout the chapter.

Chapter 2 Introduces the problem of robustness, previously studied methods and technologies, describes existing approaches to basic robustness testing, attacks simulation and defense, and explains why they are insufficient for a range of adversarial scenarios. It argues for the need of a practical and reliable robustness evaluation approach.

Chapter 3 Describes the necessary requirements that the system and its different components must satisfy in order to achieve the goals of generating adversarial examples and evaluating the robustness of deep learning models. Additionally, this chapter presents the efficiency, usability, and scalability requirements and the hardware, software, and platforms used on the project.

Chapter 4 This chapter describes the design of the system by detailing the deep learning model setup, the adversarial attack workflow, the data processing steps and how the different components of the system interact with each other. Furthermore, it explains how the Streamlit-based web interface for easy interaction and comprehension was developed.

Chapter 5 The implementation of the developed system is described in detail. First, the chosen datasets and the corresponding data preparation steps are described. This is followed by a detailed discussion of the used adversarial attack methods (FGSM and PGD) to generate robust adversarial examples. The chapter then explains how the model was evaluated, including defensive approaches and hyperparameter tuning. Moreover, the integration of the fully trained system into the user interface is described, as well as the general tools, technologies, and software libraries that were used during the implementation of the system.

Chapter 6 In this chapter, we evaluate the proposed system, discussing the obtained results, their meaning, the advantages and disadvantages of the system, as well as future plans for improvement. The main characteristics of the system are evaluated including accuracy and robustness when attacking the system. Additionally, accuracy of the system when operating in normal mode and when operating in normal and under attack modes is presented and discussed.

The References section lists all the books, research papers, websites, and other sources that were used to support and guide the development of the project.

CHAPTER 2

LITERATURE SURVEY

M. H. Meng; X. Huang; Y. Liu; B. Z. Yuan; C. Zhang[1]. Adversarial Robustness of Deep Neural Networks: A Survey from a Formal Verification Perspective. IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (T-CAD), vol. 38, no. 1, pp. 175-187, 2019. doi: 10.1109/TCAD.2018.2849025.

Advances in deep learning (DL) have led to significant improvements in various artificial intelligence (AI) applications. However, adversarial robustness and trustworthiness of deep neural networks (DNNs) have become major challenges in their real-world applications. Although many techniques and methods have been proposed to improve the robustness of well-trained models, even small perturbations to the input data can cause DNNs to behave (un)predictably. This paper provides a systematic study of existing robustness verification techniques for DNNs. We present a comprehensive analysis of current techniques, including constraint solving, abstract interpretation, and robustness verification by optimization-based methods. A taxonomy is further introduced to classify these verification approaches based on several dimensions including methodology and application scopes. Although these techniques can help formally verify whether a given DNN fulfills certain robustness properties, they still face many challenges in terms of efficiency and completeness, especially for complex models. In light of these challenges, improving efficiency and generalizing current verification techniques to a larger class of DNNs is a promising research direction in the future.

N. Carlini et al[2]. "On Evaluating Adversarial Robustness" studies by N. Carlini et al. [3] to tackle the problem "On Evaluating Adversarial Robustness". In this study the authors of the cited paper evaluate some of the current defense methods and show that many of them fail when correctly evaluated. They argue that correctly evaluating robustness is a strong and challenging task that cannot be addressed by simple (weaker) "regular" testing procedures. In this paper we discuss some of the key principles to properly evaluate robustness. We use such principles to provide a solid methodology to evaluate if a defense method really improves the robustness of a network.

We also detail best practices and guidelines that can help prevent common pitfalls during evaluation. Importantly, we observe that many current defenses appear to work because they are evaluated under very limited conditions, highlighting the potential for these methods to fail in practice. Finally, our results reinforce the importance of rigorous evaluation methods in adversarial ML, demonstrating that strong testing can both help prevent overfitting to specific (breakable) evaluation setups and guarantee the robustness of models in practice. Evaluating adversarial robustness requires specialized techniques, and this work provides strong justification for the framework that we introduced for adversarial robustness evaluation and the need for comprehensive evaluation techniques to determine the robustness of deep learning models.

N. Carlini and D. Wagner [3] present a model “Towards Evaluating the Robustness of Neural Networks”. They focus on deep neural network models and show that they are susceptible to the creation of adversarial examples. They address challenges such as the small perturbation and the need for fine-tuned experimental setups. In their study, they use multiple adversarial attacks with different approaches and distance metrics ($L0$, $L2$, $L\infty$) and test the classification behaviour of the network when it is subjected to such adversarial inputs. The study reached a high number of successfully generated adversarial examples. In particular more powerful techniques to generate adversarial examples were developed, which showed that current defenses are often not effective and reached a significantly lower error rate than previous methods.

Simple and scalable attack methods, such as input addition and swapping, were employed to test the robustness of deep learning models across a range of architectures. These attack methods were shown to be effective and easy to implement, and the authors highlighted that defensive techniques, such as defensive distillation, do not provide sufficient protection against such attacks. By deploying stronger, adaptive attacks, the study found that many commonly used defensive techniques fail to achieve any improvement in robustness, in a rigorous testing regime. This paper provides a solid case study into the vulnerabilities of adversarial examples, and highlights the importance of more effective robustness evaluation techniques for deep learning models, which are a critical component of many security-critical applications. The study provides a solid reference point for the design and evaluation of robustness evaluation techniques.

S. Anasuri [4], "Adversarial Attacks and Defenses in Deep Neural Networks". This paper presented a framework to discuss different attack and defense techniques adopted to counter adversarial examples, across a variety of deep learning architectures and applications including computer vision, speech recognition, natural language processing. The study was conducted using standard datasets, including the MNIST, CIFAR-10 and ImageNet datasets that consisted of large amounts of labelled data. Both normal and adversarial samples of data were utilised. The different types of attack included in the study were the FGSM, PGD, Carlini-Wagner and the DeepFool attacks. Additionally, adversarial training, defensive distillation and input preprocessing were the various methods used to defend against these attacks. The paper showed how these adversarial attacks can significantly affect the accuracy of a network and presented how deep neural networks can be highly unreliable in real-world scenarios. The main contribution of this paper was to give a comprehensive overview of both attacks and defenses.

This paper uses widely available tools and follows a straightforward experimentation setup to study adversarial robustness. The paper also discusses various attack setups (white-box, black-box etc.) and their effects on the neural network's predictions, and contrasts different defense strategies for increasing robustness with suitable evaluation. The paper is a seminal reference for adversarial machine learning. For our current project on adversarial robustness evaluation and defense for deep (and possibly recurrent) neural networks, this paper serves as a good starting point for understanding the current mechanisms of attacks and defenses, and for designing models that are robust and reliable.

A. Liu, Q. Gan, J. Yang, Y. Chen, X. Han, and T. Huang[5]. Training robust deep neural networks via adversarial noise propagation. IEEE Transactions on Information Forensics and Security, 12(9):2005–2016, 2017. In this paper, the authors A. Liu et al. proposed a model called "Training Robust Deep Neural Networks via Adversarial Noise Propagation". This model uses a deep neural network, improves the robustness, and uses the method of adversarial noise to train the network. Based on the experiments on MNIST, CIFAR-10 and ImageNet datasets with large amounts of high-quality images including normal images and adversarially perturbed images taken under realistic and controlled conditions, the experiments demonstrated that significant improvement in robust generalization and accuracy can be obtained by training robust deep neural networks with the proposed method. This paper used normal samples and adversarial samples collected from a variety of sources and environments to balance the “ideal world” and “real world” challenges to improve generalization ability of network and avoid overfitting. This study investigates how adding adversarial noise to hidden layers of deep neural networks can improve robustness, or resistance to attacks. We also experimentally evaluate the performance of such models in terms of accuracy, robustness to attacks, and model stability. Our results demonstrate that with proper training and defense strategy, robust models can achieve high robustness performance and even maintain accuracy in strong adversarial environments.

Chouhan et al. [6], hav X. Yuan et al. proposed a improved model “A Simple Framework to Enhance the Adversarial Robustness of Deep Learning-based Intrusion Detection System” in [6]. The improved model is designed to enhance robustness besides the common evaluation metrics by introducing both detection and classification methods that are capable of resisting adversarial inputs. The improved model utilizes deep learning to extract complex features relevant to network traffic and learn effective detection functions from massive data.

Meanwhile, a secondary improved machine learning-based detection approach is introduced, which utilizes local intrinsic dimensionality (LID) and other methods, and then constructs a parallel decision-making structure to fuse the improved deep learning-based model and the improved machine learning-based model at the decision level. Experimental results on four benchmark datasets demonstrate that the improved model has better overall accuracy and improved robustness against attacks.

Feature Normalization Methods for Detecting Adversarial Examples. This paper further explores how pre-processing methods can complement the learned representations of deep learning models to detect adversarial examples. In particular, it examines how a hybrid of a deep learning model and feature normalization/ noise removal methods can distinguish between clean inputs and adversarial samples generated with the intent to deceive current deep learning models. The paper describes several experiments to demonstrate the state-of-the-art performance of the hybrid system on image classification tasks.

ClassificY. Li et al[7]. propose a model titled “Towards Robustness of Deep Neural Networks via Regularization.” Using deep learning approach, the model enhances adversarial robustness and incorporates regularized embedded space to control influence of adversarial samples. In the experiments, the group used a number of standard image classification datasets (including CIFAR-10 and ImageNet), as well as samples of different class distributions under regular and adversarial settings. The paper trains a typical neural network with embedding regularization, as well as a model that incorporated sample detection techniques to distinguish between regular and adversarial samples. Results showed that the model delivered improved robustness when trained using regularized embeddings and adversarial examples. One of the key contributions in this work is pushing adversarial samples back in the distribution of the training data, which is crucial in building robust and stable models capable of delivering reliable results.

The paper also extensively explored preprocessing and feature transformation methods, including how dimensionality reduction and feature normalization affect robustness-achievability. It also demonstrated that a well-crafted neural network together with embedding regularization and training strategy improves robustness significantly, and such robust models can be applied to practical systems where security is a large concern.

Goodfellow et al. [8], in their paper “Explaining and Harnessing Adversarial Examples”, first introduced the FGSM (Fast Gradient Sign Method) which was a turning point in how people perceive deep learning models and their robustness. They highlighted the sensitivity of deep neural networks and Lack of Robustness, and proposed a method to generate adversarial samples. In this paper, we explain how Fast Gradient Sign Method works, and how it generates adversarial examples. FGSM uses gradient-based optimization, and introduces little noise to input data. As a result, the generated adversarial examples almost imperceptible to human eyes, yet can cause deep learning models to misclassify. The technique uses gradient sign perturbations, which greatly reduces the computational cost in generating adversarial samples. Extensive experiments have been performed to evaluate the efficacy of FGSM, by testing on various benchmark datasets. The results indicate high success rates in fooling a wide range of neural networks, while outperforming strong existing attack methods with this innovation, systematic studies could now be performed on the vulnerability of deep learning models in adversarial situations. Such scenarios are critical to study in applications such as autonomous cars and healthcare. The paper inspired the concept of using adversarial training as a defense mechanism. While this paper does not address robustness evaluation frameworks, it laid the foundation for many attack and defense techniques developed later. In this paper, the founder is particularly interested in illustrating the vulnerabilities of deep learning models, but the work ultimately contributes to the development of stronger and more robust models that are critical for developing safe and trustworthy AI models. In the current adversarial robustness project, gradient-based attack methods like FGSM and PGD are being used to test the security of deep learning models.

A. Liu, Y. Sun, H. Tang, C. C. Aggarwal, Z.-H. Zhou[9], Training Robust Deep Neural Networks via Adversarial Noise Propagation, In International Conference on Computational Intelligence, In Coimbra, Portugal, 2019. [Download(s): 22] ÂSTRACT We explore the possibilities of advanced adversarial defense techniques in deep learning, in particular, one of the sections of this study focuses on training robust deep networks that are able to prevent both adversarial examples and noisy inputs. We present a systematic framework, DNN-based, to tackle robustness problem by taking a unique perspective in addressing robustness: i) data-level and ii) model-level. Within this framework, we focus on learning to protect deep networks from adversarial attacks through robust overfitting by exploring how to embed robust generalization capability into learned models. We conduct extensive experiments on several benchmark datasets with diverse settings and achieve consistently better robust error rates and accuracy than existing works depending on attack methods and model configurations. Specifically, by injecting adversarial noise into the training process, we not only introduce perturbations into inputs but also propagate noise within hidden layers during training.

To improve the robustness of the network, we employ various techniques including mixup, batch normalization and regularization to stabilize training and prevent overfitting. We further evaluate the robustness of the models with robustness accuracy, and various attack success rates. We compare the robustness of different models under different types of attacks, and analyze the trade-off between accuracy and robustness.

X. Yuan et al. in [10], “A Simple Framework to Enhance the Adversarial Robustness of Deep Learning-based Intrusion Detection System,” presented a hybrid deep learning and machine learning model to effectively distinguish adversarial samples from normal network traffic. In this work, deep neural networks were utilized for complex feature extraction from data samples and then the data samples were inputted into a machine learning model to obtain the final classification result. To improve the learning accuracy of the model, the collected network traffic data set after the expansion was preprocessed including feature value normalization, noise filtering, and feature dimension reduction. Based on the processed features, a deep learning model consisting of multiple layers with activation functions to extract features, was designed to generate features and then these obtained features were inputted into a machine learning model for classification purposes.

Combining models improves robustness and accuracy compared to individual deep learning models as well as classic machine learning methods in adversarial scenarios. They achieve higher reliability and detection accuracy while individual deep learning models learn better features. The achieved results are further visualized by accuracy, attack detection rate and confusion matrices. Importantly, the hybrid model strikes a good balance between robustness and efficiency for real-time system needs. This study provides the first experience of applying adversarial robustness evaluation and defense systems using hybrid models, and is particularly relevant for systems that demand high security and consistency in practice.

Salman Raza et al. proposed a novel framework named RNAF[11] (RNAF: ResilienceNet Adversarial Framework Using DRL for Adversarial Attacks on Digital Images).The proposed framework, RNAF, can fortify the security of deep learning-based image classification systems against various types of adversarial attacks by using a combination of Deep Reinforcement Learning (DRL), adversarial training, and adaptive defense strategies to detect input images that have been manipulated in order to mislead a system. The system continuously learns from the various types of attacks and improves the detection by fine-tuning the policies in real time to counter the newly introduced types of attacks. To detect hidden malicious modifications in digital images, RNAF incorporates a number of techniques, including feature extraction, perturbation analysis, and image reconstruction. Experimental results show that the framework significantly improves the classification accuracy and decreases the attack success rate and number of false predictions when tested on CIFAR-10 and ImageNet datasets by using a number of attacks including FGSM, BIM, PGD, and CW attacks. The results also demonstrate that the proposed framework outperforms the state-of-the-art CNN and existing defense techniques. Therefore, by integrating DRL with various types of adversarial attack defense techniques, RNAF can make deep learning-based image security systems more robust and reliable.

Survey “Adversarial Robustness of Deep Learning: Theory, Algorithms, and Applications” by Wenjie Ruan et al.[12].In this survey, the authors, Wenjie Ruan et al. first explained the adversarial machine learning and the vulnerabilities of deep neural networks in details. They explained that adding a small perturbation to input could mislead the deep learning models to output incorrect results. After that, the authors described different types of attacks including evasion attacks, poisoning attacks, white-box attacks, black-box attacks, and transfer-based attacks. Furthermore, they explained several methods for defending against adversarial attacks including adversarial training, defensive distillation, certified robustness verification, gradient masking, and feature squeezing methods. The authors also introduced several theoretical robustness analysis methods for measuring the stability of neural networks. The authors also described several applications, including autonomous vehicles, healthcare systems, cyber security, face recognition, and speech processing. Finally, experimental results showed that by combining different methods for enhancing robustness, several methods can improve the robustness and the trustworthiness of AI systems against adversarial attacks.

Mamoun Alazab, MingJian[13] Tangpaper analyzes the Deep Learning models such as CNN, RNN, Autoencoders, GANs and Deep Belief Networks and their Deep Learning Applications for Cyber Security. These models can be used for the detection of cyber threats and malicious activities. The applications of Deep Learning models are malware classification, intrusion detection and prevention system, phishing attack detection, spam filtering, anomaly detection, botnet detection and network traffic analysis. Deep Learning models automatically learn the hidden patterns within the large datasets of Cyber Security without requiring the feature extraction. The paper states that the Deep Learning systems are capable of handling large amounts of both the structured and unstructured data. The experimental results of Deep Learning models on the various datasets of Cyber Security provide the comparison with traditional Machine Learning models. The results show that the Deep Learning models provide better detection accuracy, less number of false alarms, faster detection time and ability to detect evolving attacks.

X. Yuan et al[14]. in their paper titled “A Defensive Framework Against Adversarial Attacks on ML-Based NIDS” proposes an improved framework for securing the Machine Learning-based Network Intrusion Detection Systems (NIDS) against the Adversarial Attacks.This framework aims to enhance the security of ML-based NIDSs against adverse impacts caused by ML attacks. To achieve this objective, an improved framework consists of a primary part which is based on a deep learning-based IDS and a secondary part based on a machine learning-based adverse impact detector (AID) such as LID. The decisions from both parts are fused in a parallel decision fusion manner at the decision level to enhance attack detection performance. The framework can effectively distinguish between attacks and normal network traffic while maintaining a high classification rate for both attacks and normal network traffic. In addition, a number of preprocessing steps and feature normalization approaches are introduced in the framework to counter malicious perturbations that affect the performance of learning from network traffic data. Experiments are conducted on a number of datasets such as NSL-KDD and CICIDS using various types of attacks to test the framework’s robustness. The results reveal that the improved framework is capable of achieving a number of advantages compared to other IDSs, including improved robustness, higher detection rates, and lower false positive rates while providing good resistance to adverse impacts caused by generated adverse samples.

Kui Ren Tianhang Zheng ,Zhan Qin ,Xue Liu[15] paper, “Adversarial Attack, Defense, and Applications with Deep Learning Frameworks” by the researchers provides a comprehensive review on the adversarial machine learning techniques and its respective defense techniques.Adversarial examples are generated by adding noise to inputs in order to deceive DL models. Several types of attack strategies such as FGSM, PGD, DeepFool, Carlini-Wagner (CW) attack, black-box attack and transfer attack are explained in the paper. Also, adversarial training, feature normalization, gradient masking, defensive distillation, noise filtering, input preprocessing, etc. as attacks’ defense strategies are explained. Hybrid models that combine input preprocessing with DL models improve adversarial robustness greatly. The applications of adversarial learning are image classification, cyber security, health care, self-driving car, biometric, speech recognition, etc. Experimental results show that robust defense mechanisms decrease the attack success rate and increase the classification accuracy, thus improve the reliability and security of corresponding applications greatly.

2.1 Summary of Literature Survey

The survey of literature represented in Table 2.1 discusses various deep learning and adversarial learning methods developed for detecting, analyzing, and improving the robustness of AI models against adversarial attacks, highlighting the techniques used, attack and defense mechanisms applied, and the performance outcomes achieved under different testing conditions.

Table 2.1:Summary of Literature Survey

No	Author	Title	Features	Pros	Cons
1	Charis Eleftheriadis,Panagiotis Katsaros	Adversarial Robustness Improvement for Deep Neural Networks	Uses adversarial training with different attack examples to improve neural network robustness.	Enhances security and robustness of deep learning models against attacks.	Requires high computational power and increased training time.
2	Xinwei Yuan, Shu Han, Wei Huang, Hongliang Ye	A Simple Framework to Enhance the Adversarial Robustness Deep Learning-based Intrusion Detection System	Combines Deep Learning and Machine Learning models for attack detection and adaptive decision making.	Improves intrusion detection accuracy under adversarial attacks.	Frameworkcomplexity increases due to the use of multiple models.
3	Sunil Anasuri	Adversarial Attacks and Defenses in Deep Neural Networks	Explains different adversarial attack and defense techniques using popular datasets.	Provides a broad understanding of attack and defense mechanisms.	Mainly theoretical; limited practical implementation details.
4	Mark Huasong Meng, Guangdong Bai	Adversarial Robustness of Deep Neural Networks: A Survey from	Discusses formal verification methods to evaluate neural network safety	Identifies research gaps and improves understanding of model verification.	Formal verification methods are computationally

		a Formal Verification Perspective	and robustness.		expensive and difficult to scale.
5	Jun Guo, Wei Bao, Jiakai Wang, Yuqing Ma	Comprehensive Evaluation Framework for Deep Model Robustness	Uses multiple methods to evaluate robustness of deep learning models and defenses.	Provides a structured framework for comparing model robustness.	Evaluation process can be complex and time-consuming.
6	Yao Li, Martin Renqiang Min, Thomas Lee	Towards Robustness of Deep Neural Networks through Regularization	Uses regularization techniques to reduce noise effects and improve robustness.	Enhances resistance against strong adversarial attacks.	May reduce model accuracy on normal clean data.
7	Aishan Liu, Xianglong Liu, Chongzhi Zhang	Training Robust Deep Neural Networks through Adversarial Noise Propagation	Introduces adversarial noise during training to strengthen models.	Improves performance on corrupted and attacked images.	Training becomes computationally expensive.
8	Samuel Henrique Silva, Peyman Najafirad	Opportunities and Challenges in Deep Learning Adversarial Robustness	Reviews defense techniques, challenges, and future research directions.	Provides comparative analysis of different robustness methods.	Mostly survey-based with limited experimental validation.
9	Nicholas Carlini, Anish Athalye, Nicolas Papernot	On Evaluating Adversarial Robustness	Explains proper evaluation methods and highlights testing mistakes.	Encourages accurate robustness evaluation using strong attacks.	Focuses more on evaluation than on new defense techniques.
10	Ziang Yan, Yiwen Guo, Changshui Zhang	Deep Defense: Training DNNs with the Improved Adversarial Robustness	Adds additional protection during training to resist adversarial perturbations	Reduces harmful perturbation effects and improves robustness.	Increased training complexity and resource usage.
11	Salman Raza et al.	RNAF: ResilienceNet Adversarial Framework	Uses DRL for adversarial image defense.	Improves robustness against attacks.	High computational cost.

		Using DRL for Adversarial Attacks on Digital Images			
12	Wenjie Ruan et al.	Adversarial Robustness of Deep Learning: Theory, Algorithms, and Applications	Explains attack and defense methods.	Improves model reliability.	Complex robustness techniques.
13	Mamoun Alazab, MingJian Tang	Deep Learning Applications for Cyber Security	Uses deep learning for cyber threat detection.	Better detection accuracy.	Needs large datasets.
14	Xinwei Yuan et al.	A Defensive Framework Against Adversarial Attacks on ML-Based NIDS	Combines DL and ML for intrusion detection.	Improves attack detection.	Complex framework design.
15	Kui Ren et al.	Adversarial Attack, Defense, and Applications with Deep Learning Frameworks	Reviews adversarial attacks and defenses.	Comprehensive study of methods.	Limited practical implementation.